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(54) **CAMERA MODULE AND TERMINAL DEVICE**

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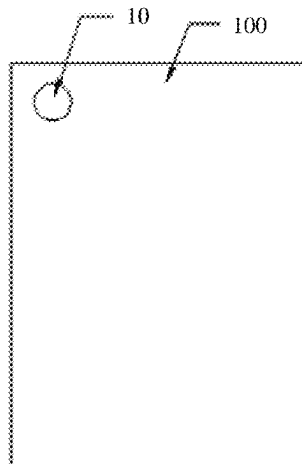
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(57) **ABSTRACT**
A camera module and a terminal device having the camera module, the camera module including a plurality of lenses, where lenses of the plurality of lenses are sequentially arranged from an object side to an image side along a direction of an optical axis, where at least one of the plurality of lenses is a free-form lens, where the free-form lens is a non-rotationally symmetric lens, where a first lens of the plurality of lenses is a lens of the plurality of lenses nearest the object side in a direction from the object side to the image side, where a distance on the optical axis between an object-side surface of the first lens and an imaging surface is TTL, where an effective focal length of the camera module is EFL, and $TTL/EFL \leq 2.0$.

19 Claims, 10 Drawing Sheets



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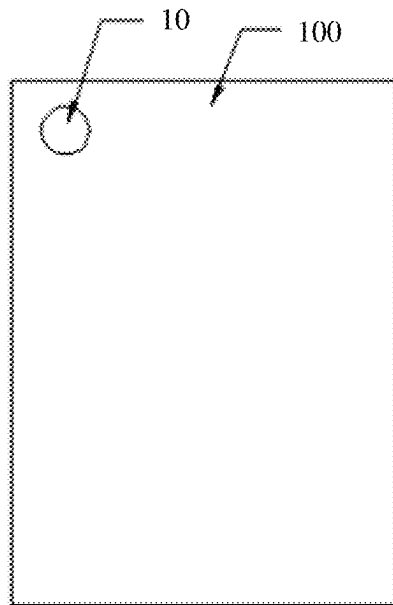


FIG. 1

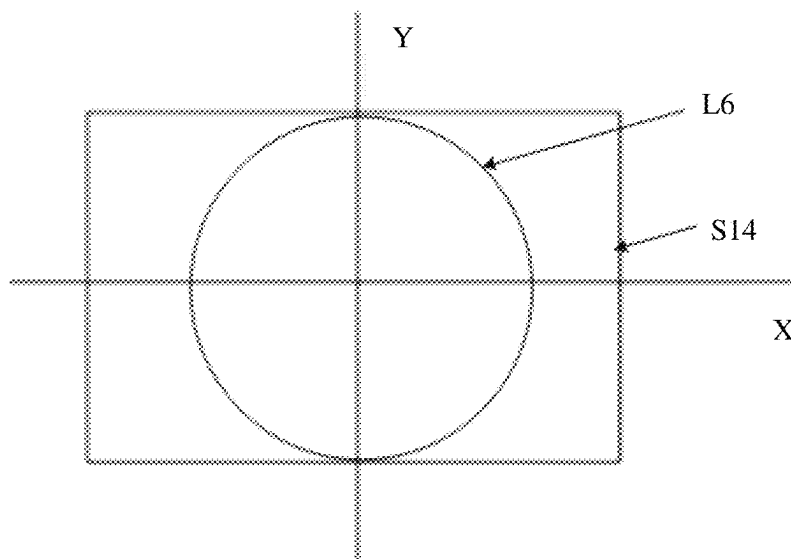


FIG. 1a

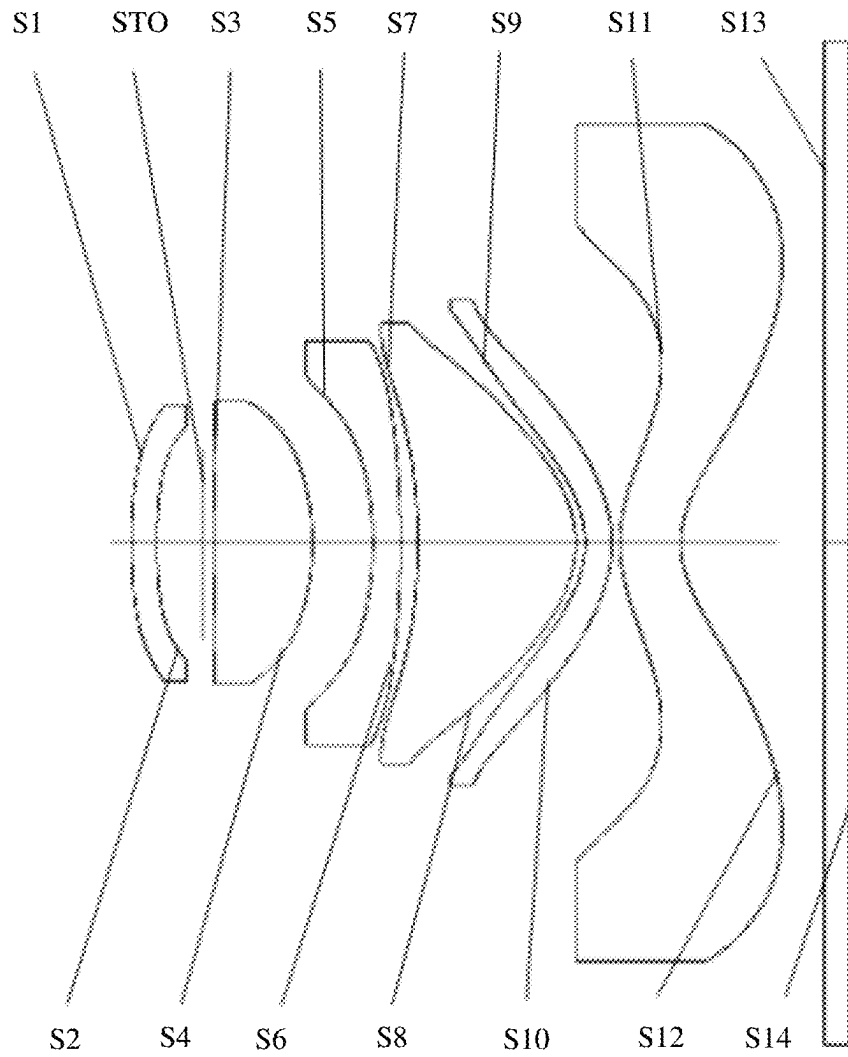


FIG. 2a

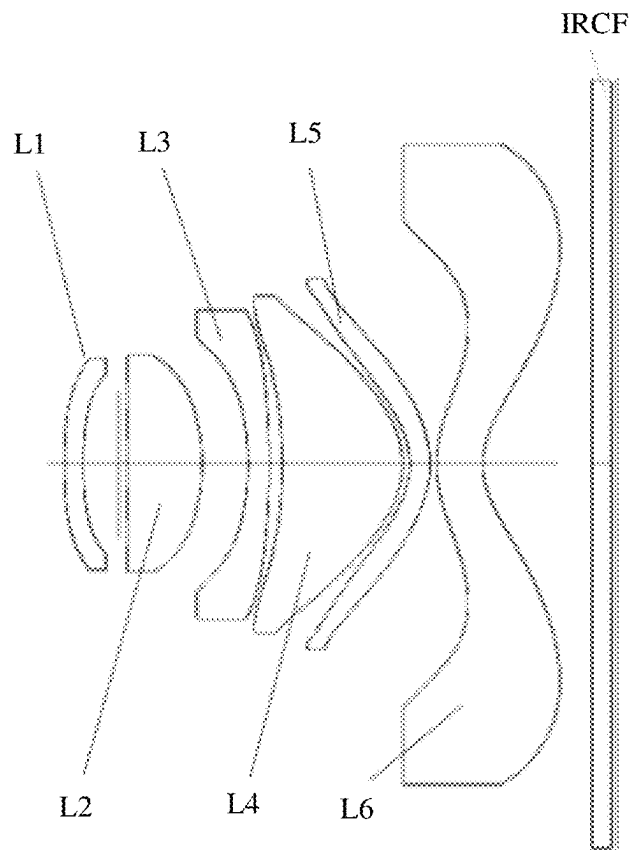


FIG. 2b

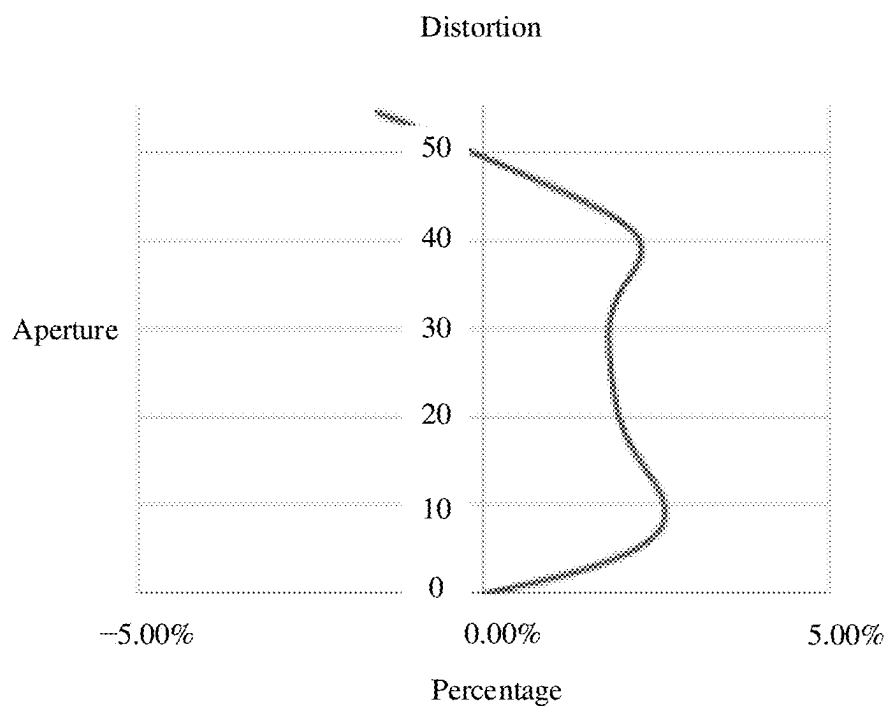


FIG. 2c

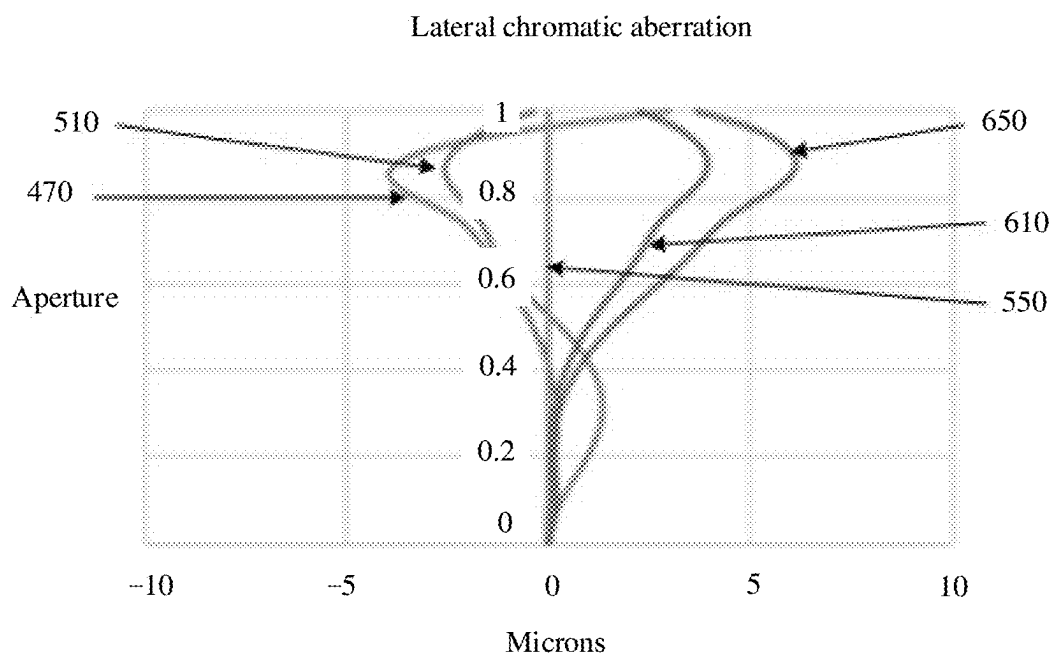


FIG. 2d

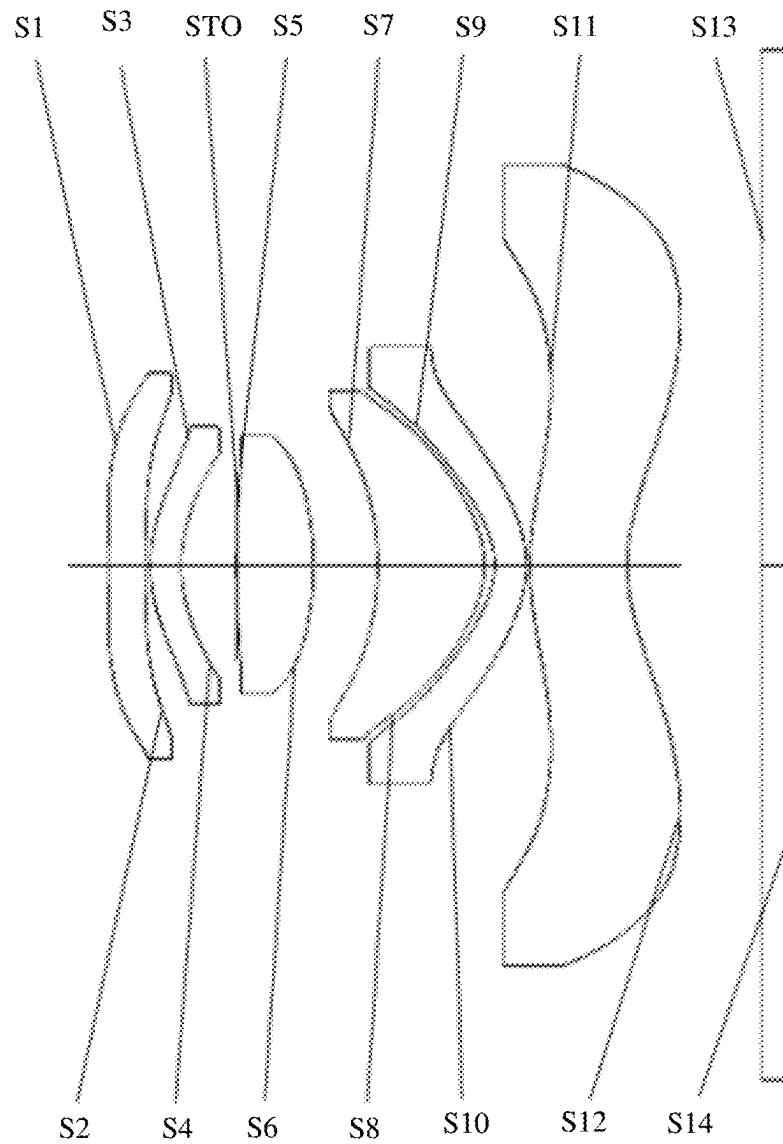


FIG. 3a

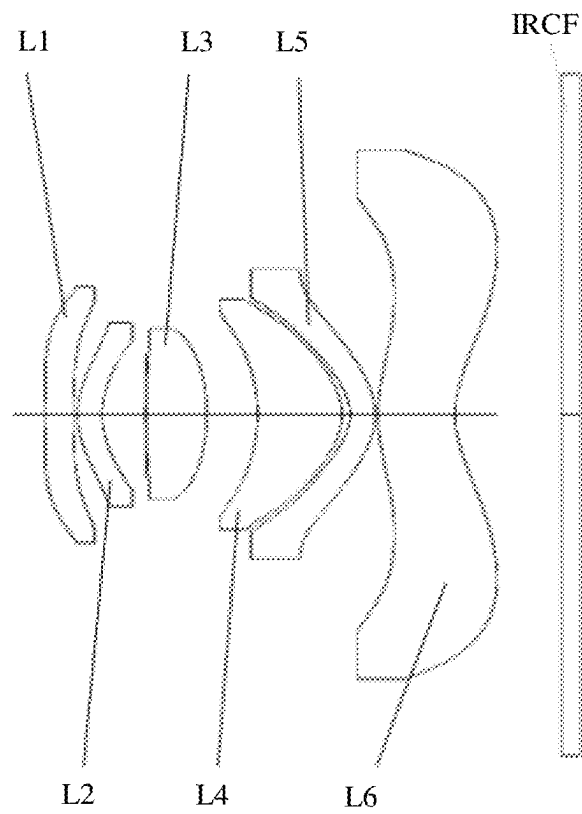


FIG. 3b

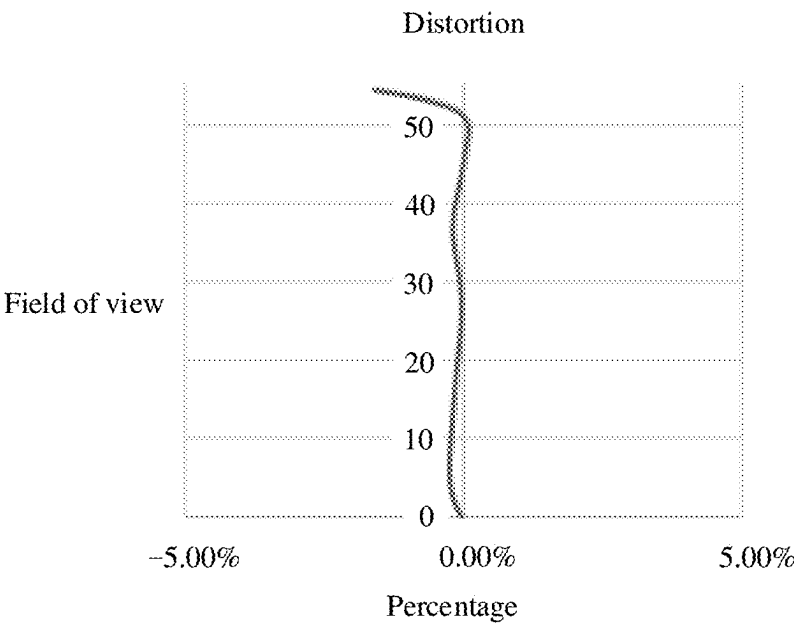


FIG. 3c

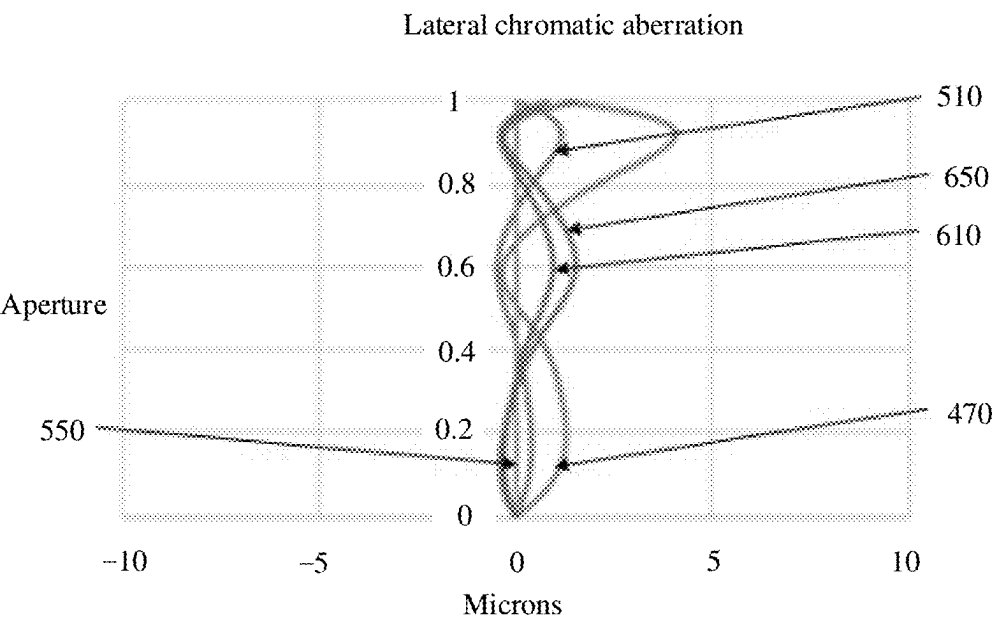


FIG. 3d

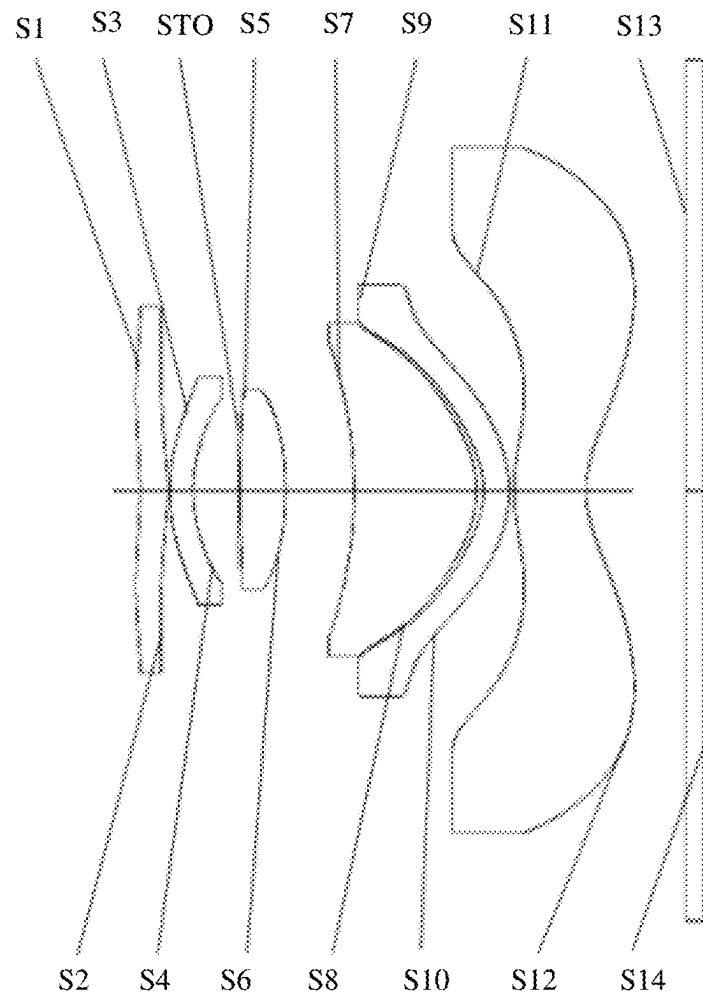


FIG. 4a

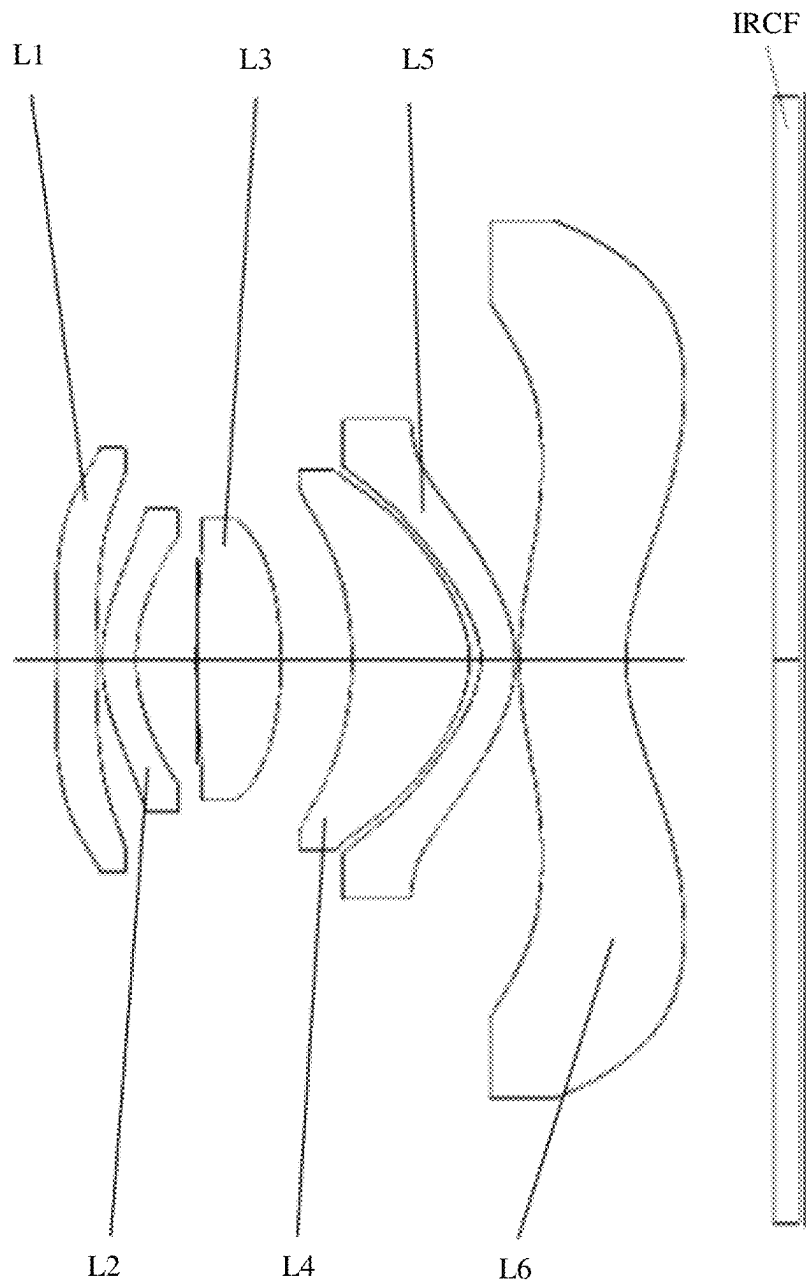


FIG. 4b

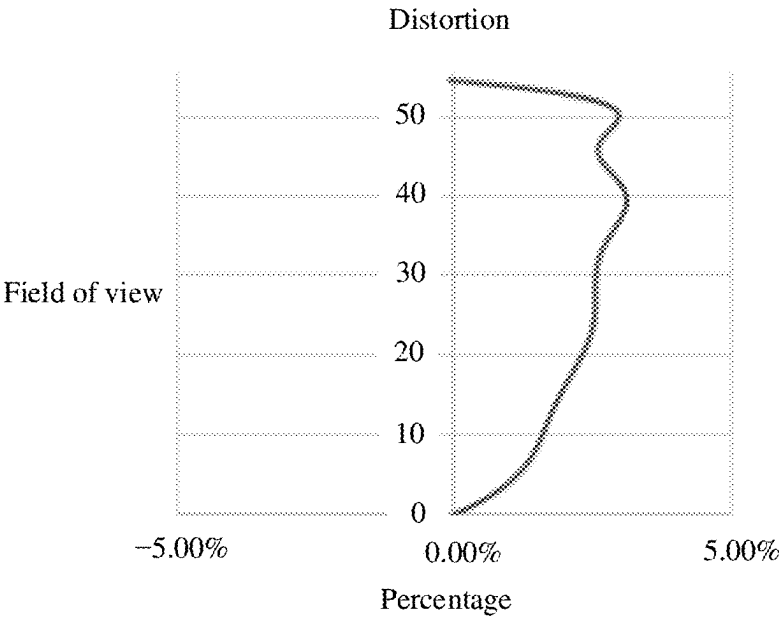


FIG. 4c

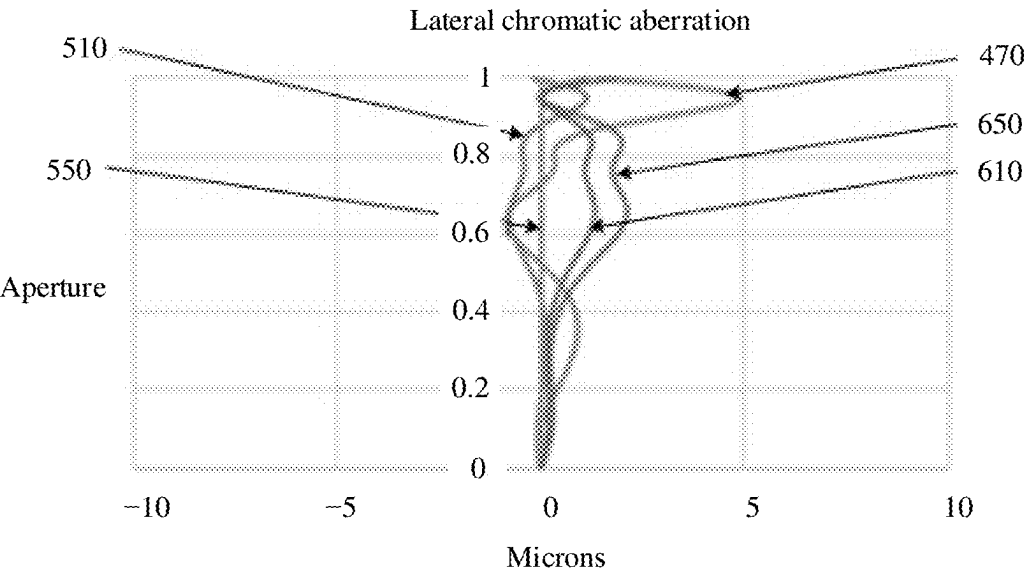


FIG. 4d

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CAMERA MODULE AND TERMINAL
DEVICECROSS REFERENCE TO RELATED
APPLICATIONS

This application is a national stage of International Application No. PCT/CN2020/106958, filed on Aug. 5, 2020, which claims priority to Chinese Patent Application No. 201911205850.4, filed on Nov. 29, 2019 and Chinese Patent Application No. 201910734247.9, filed on Aug. 9, 2019. All of the aforementioned applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

This application belongs to the field of optical imaging technologies, and in particular, to a camera module and a terminal device of a photographic optical system.

BACKGROUND

With the popularity of mobile terminal devices, photography technologies for users to obtain images have developed rapidly. To meet a wide range of market needs, large sensor sizes, large apertures, a diversity of fields of view, and compact structures are important development trends for camera modules. From single-camera modules, to dual-camera modules, and to multi-camera modules, a combination of a plurality of field-of-view lenses has been implemented. This also allows ultra wide-angle lens sets to be applied to mobile terminal devices.

Ultra wide-angle lenses have larger fields of view, but it is difficult to balance a contradiction between an optical distortion and a compact structure. Therefore, most ultra wide-angle lenses have relatively large optical distortions, and a TV distortion of an image is also very obvious. In the field of mobile electronic camera modules limited by compact structures, an optical distortion problem is more prominent and is difficult to resolve. In addition, in application of terminal devices, during video recording by using an ultra wide-angle lens, real-time distortion correction for a video image consumes a large quantity of processing resources and is difficult to achieve.

How to resolve the optical distortion problem of the ultra wide-angle lens set should be a direction of research and development in the industry.

SUMMARY

Embodiments of this application provide a camera module and a terminal device. The camera module is an ultra wide-angle lens, and a non-rotationally symmetric free-form lens is introduced into the lens, to resolve an optical distortion problem of the ultra wide-angle lens, implement an ultra wide-angle and low-distortion imaging effect, and provide a user with good experience.

According to a first aspect, this embodiment provides a camera module, applied to a terminal device, and includes a plurality of lenses sequentially arranged from an object side to an image side along a direction of an optical axis. A quantity of the lenses may be three, four, five, six, seven, or the like. At least one lens of the plurality of lenses is a free-form lens. The free-form lens is a non-rotationally symmetric lens, the 1st lens in the plurality of lenses in a direction from the object side to the image side is a first lens, a distance on the optical axis between an object-side surface

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of the first lens and an imaging surface is TTL, an effective focal length of the camera module is EFL, and TTL/EFL≤2.0, to implement relatively short TTL. The distance on the optical axis between the object side surface of the first lens and the imaging surface is TTL, which is specifically a distance between an intersection of an object-side surface of the first lens and the optical axis and an intersection of the imaging surface and the optical axis. In this application a surface type of the free-form lens is non-rotationally symmetric. The non-rotationally symmetric free-form lens can increase freedom of optical design of the camera module, and a rectangular imaging region can be implemented without limiting an imaging region to being a rotationally symmetric image circle. An electronic image sensor is arranged on the imaging surface, and is specifically a sensor chip of a camera in the terminal device. In this application, at least one lens is limited to being a free-form lens, to suppress an optical distortion problem of the camera module. An imaging effect of the camera module can be ensured even in the case of an ultra wide angle. In addition, a relatively short total length of the camera module can be obtained. In other words, the non-rotationally symmetric free-form lens is introduced into the camera module, so that an ultra wide-angle and low-distortion imaging effect can be implemented, that is, a system aberration of the camera module can be reduced or minimized, to implement functions of aberration correction and distortion reduction. In addition, the free-form lens can further reduce TTL of the camera module, so that the camera module has a compact structure.

In a possible implementation, an X-axis and the optical axis form a first plane, a Y-axis and the optical axis form a second plane, and the X-axis and the Y-axis are two central axes that are perpendicular to each other on the imaging surface of the camera module. An intersection of the X-axis and the Y-axis is located on the optical axis. The free-form lens is a centrosymmetric structure by using the first plane as a center, and the free-form lens is also a centrosymmetric structure by using the second plane as a center. Symmetry of the free-form lens in a direction of the X-axis and in a direction of the Y-axis helps ensure imaging quality. The imaging region of the electronic sensor is rectangular. Symmetry of the free-form lens in the direction of the X-axis and in the direction of the Y-axis makes imaging quality of the imaging region keep particular symmetry. This helps implement that imaging quality of a middle region close to the optical axis is better than imaging quality of an edge region away from the optical axis.

In a possible implementation, an object-side surface or an image-side surface of the free-form lens or both are free-form surfaces, and a surface type expression of the free-form surface is:

$$z = \frac{cr^2}{1 + \sqrt{1 - (1 + k)c^2 r^2}} + \sum_{i=1}^M A_i E_i$$

z is a sag of an optical surface, and z is an expression of x and y; k is a conic coefficient; c is a curvature radius; r is a radius height in the direction of the optical axis; $r^2 = x^2 + y^2$; A_i is a polynomial coefficient; and E_i is a monomial of an X-axis coordinate and a Y-axis coordinate. Directions of the x-axis and the y-axis herein are consistent with the foregoing directions of the X-axis and the Y-axis about which the free-form lens is symmetric.

$$\sum_{i=1}^M A_i E_i = A_1 x^0 y^2 + A_2 x^2 y^0 + A_3 x^2 y^2 +$$

$$A_4 x^4 y^0 + A_5 x^0 y^4 + A_6 x^6 y^0 + A_7 x^0 y^6 + A_8 x^4 y^2 + A_9 x^2 y^4 + \dots$$

Exponents of x and y in E_i of the surface type expression of the free-form lens are both even numbers, x is an X-axis coordinate, and y is a Y-axis coordinate, so that the surface type of the free-form lens has symmetry. Specifically, the exponents of x and y in E_i of the surface type expression of free-form lens are both even numbers. This can make the surface type of the free-form lens better symmetric, and facilitate lens processing and detection.

In a second implementation, an object-side surface or an image-side surface of the free-form lens or both are free-form surfaces, and a surface type expression of the free-form lens is:

$$z = \frac{cr^2}{1 + \sqrt{1 - (1+k)c^2 r^2}} + \sum_{i=1}^M A_i E_i$$

z is a sag of an optical surface, and z is an expression of x and y; k is a conic coefficient; c is a curvature radius; r is a radius height in the direction of the optical axis; $r^2 = x^2 + y^2$; A_i is a polynomial coefficient; and E_i is a monomial of an X-axis coordinate and a Y-axis coordinate.

$$\sum_{i=1}^M A_i E_i = A_1 |x^1 y^0| + A_2 |x^0 y^1| + A_3 |x^1 y^1| + A_4 |x^0 y^2| +$$

$$A_5 |x^3 y^0| + A_6 |x^2 y^1| + A_7 |x^1 y^2| + A_8 |x^0 y^3| + A_9 |x^4 y^0| + \dots$$

A_i is a polynomial coefficient, x is an X-axis coordinate, and y is a Y-axis coordinate.

In a third implementation, an object-side surface or an image-side surface of the free-form lens or both are free-form surfaces, and a surface type expression of the free-form lens is:

$$z = \frac{c_x x^2 + c_y y^2}{1 + \sqrt{1 - (1+k_x)c_x^2 x^2 - (1+k_y)c_y^2 y^2}} + \sum_{i=1}^M A_i |x^i| + \sum_{j=1}^M B_j |y^j|$$

z is a sag of an optical surface, and z is an expression of x and y; x is an x-axis coordinate, and y is a Y-axis coordinate; k_x and k_y are conic coefficients; c_x and c_y are curvature radii; and A_i and B_j are polynomial coefficients.

In a possible implementation, the X-axis and the Y-axis are two central axes passing through a center of the imaging surface and respectively parallel to a long side and a short side of the imaging surface.

In a possible implementation, a quantity of the plurality of lenses is N, $N \geq 3$, and the plurality of lenses include the first lens to the N^{th} lens sequentially arranged in the direction from the object side to the image side; and surface types of object-side surfaces and image-side surfaces of the first lens to the $(N-1)^{th}$ lens are all aspheric surfaces, and the N^{th} lens is a free-form lens. A quantity of lenses used in an embodiment of this application is six. In another implementation, the quantity of lenses may be seven, eight, or the like. In an

actual implementation process, a camera module including three, four, or five lenses may be disposed as required. The free-form lens is introduced into the camera module, so that a visual effect without image distortion is achieved for an ultra wide-angle lens set, and imaging quality unachievable by ordinary aspheric optical design can be achieved.

In a possible implementation, a half of a diagonal length of an effective pixel region of the imaging surface of the camera module is ImgH , and $\text{TTL}/\text{ImgH} \leq 2.0$, to help limit a total system length of the camera module, so that the camera module has a compact structure and meets a design requirement for a portable device.

In a possible implementation, an entrance pupil diameter of the camera module is EPD, and $\text{EFL}/\text{EPD} \leq 2.2$. In this implementation, $\text{EPD}/\text{EPD} \leq 2.2$ is used for limitation, so that an f-number of the camera module is limited, which is conducive to imaging quality.

In a possible implementation, a field of view of the camera module is FOV, $\text{FOV} \geq 100^\circ$, and $\text{EFL} < 20 \text{ mm}$.

The camera module has an ultra wide-angle imaging effect. In this implementation, the free-form lens is introduced into the camera module, to achieve a short-focal-length, large-field-of-view, and low-distortion imaging effect for the ultra wide-angle lens set. In addition, a relatively short total length of the camera module is obtained, and a compact structure of the ultra wide-angle lens set is implemented.

In a possible implementation, the quantity of the plurality of lenses is N, and the first three lenses arranged sequentially in the direction from the object side to the image side are respectively the first lens, a second lens, and a third lens. The camera module further includes a vignetting stop, and the vignetting stop is disposed on an object side of the second lens or on an object side of the third lens. The vignetting stop can intercept an imaging ray with a large aberration (also referred to as intercepting a ray of a point beyond the axis), to improve imaging quality. The vignetting stop is arranged close to a middle position of the camera module, to help balance a comprehensive aberration of the camera module.

In a possible implementation, the lens adjacent to the imaging surface in the plurality of lenses is the free-form lens, a curvature radius of the object-side surface of the free-form lens is R61, and a curvature radius of the image-side surface of the free-form lens is R62. The following condition is met: $|f/R61| + |f/R62| < 2$, to help correct the comprehensive aberration of the camera set. When $|f/R61| + |f/R62|$ is greater than or equal to 2, it does not help correct the comprehensive aberration of the camera module, and an imaging effect is affected.

In a possible implementation, the camera module further includes an infrared filter element located between the plurality of lenses and the imaging surface. The infrared filter element can effectively alleviate color cast generated around an image.

Optical design of the free-form lens can reduce the quantity of lenses of the camera module, and decrease a weight of the camera module, thereby facilitating a lightweight implementation. A flexible spatial layout and design freedom of the free-form lens simplify a structure of the camera module. The optical design of the free-form lens increases freedom of optimization, and helps control the system aberration of the camera module and improve overall quality of the camera module.

In a possible implementation, the camera module further includes an electronic image sensor, the electronic image sensor is disposed on the imaging surface, and the imaging surface of the camera module is a rectangular region, which matches an image sensing area of the electronic image

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sensor and is not less than the image sensing area of the electronic image sensor. In other words, the electronic image sensor is rectangular, and the imaging surface of the camera module using the free-form lens is a rectangular region, to match the image sensor, and improve imaging quality.

In a possible implementation, a diagonal length of the image sensing surface of the electronic image sensor is not less than 5.5 mm. For image sensors with same effective pixels, generally, a larger size of an image sensor indicates a larger unit area of each pixel, higher performance, more image details that can be recorded. In this implementation, the diagonal length of the image sensing surface is limited to being not less than 5.5 mm. This can ensure that the camera module applied to a mobile terminal has high image sensing performance, and provides high image quality.

According to a second aspect, this application provides a terminal device, including the camera module according to any one of the foregoing implementations.

In this application, the non-rotationally symmetric free-form lens is introduced into the lens set, to resolve an optical distortion problem of the ultra wide-angle lens set, eliminate impact of imaging distortion of the camera module, improve user experience, and obtain excellent optical quality.

BRIEF DESCRIPTION OF THE DRAWINGS

To describe the technical solutions in the embodiments of this application or the background more clearly, the following describes the accompanying drawings required for use in the embodiments of this application or the background.

FIG. 1 is a schematic diagram of a camera module applied to a terminal device according to this application;

FIG. 1a is a schematic diagram of a free-form lens that has line symmetry in a direction of an X-axis and in a direction of a Y-axis;

FIG. 2a and FIG. 2b are schematic diagrams of a camera module according to Embodiment 1 of this application;

FIG. 2C is a distortion curve of an optical system in Embodiment 1;

FIG. 2d is a lateral chromatic aberration curve of an optical system in Embodiment 1;

FIG. 3a and FIG. 3b are schematic diagrams of a camera module according to Embodiment 2 of this application;

FIG. 3c is a distortion curve of an optical system in Embodiment 2;

FIG. 3d is a lateral chromatic aberration curve of an optical system in Embodiment 2;

FIG. 4a and FIG. 4b are schematic diagrams of a camera module according to Embodiment 3 of this application;

FIG. 4c is a distortion curve of an optical system in Embodiment 3; and

FIG. 4d is a lateral chromatic aberration curve of an optical system in Embodiment 3.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

The following describes the embodiments of this application with reference to the accompanying drawings in the embodiments of this application.

Refer to FIG. 1. A camera module 10 in this application is applied to a terminal device 100. The terminal device 100 may be a portable terminal such as a mobile phone or a tablet, and the camera module 10 may be an ultra wide-angle lens set. The camera module 10 is assembled inside the terminal device 100, and may be a rear camera or a front

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camera of the terminal device 100, or a retractable camera that may extend out of a housing of the terminal device 100.

In an implementation, the camera module provided in this application includes six lenses (six lenses are used as a specific embodiment for description, and a quantity of lenses is not limited in this application). The six lenses are sequentially distributed from an object side to an image side along a direction of an optical axis as follows: a first lens, a second lens, a third lens, a fourth lens, a fifth lens, and a sixth lens. The quantity of the lenses is not limited to six, and may be another quantity, such as three, four, five, seven. At least one lens of the plurality of lenses is a free-form lens, and the free-form lens is a non-rotationally symmetric surface type. The non-rotationally symmetric free-form lens can increase freedom of optical design of the camera module. A distance on the optical axis between an object-side surface of the first lens and an imaging surface is TTL, an effective focal length of the camera module is EFL, and $TTL/EFL \leq 2.0$, to implement relatively short TTL, thereby facilitating a compact structure of the camera module. At least one lens is limited to being a non-rotationally symmetric free-form surface, so that an optical distortion problem of the ultra wide-angle lens set can be alleviated, and the camera module has a compact structure, thereby improving user experience. Aspheric curve equation of lenses 1 to 5 is as follows:

$$z = \frac{cr^2}{1 + \sqrt{1 - (1+k)c^2r^2}} + \sum_{i=1}^M \alpha_i \rho_i^i$$

z is a sag of an optical surface, and z is an expression of x and y; k is a conic coefficient; c is a curvature radius; r is a radius height in the direction of the optical axis; $r^2 = x^2 + y^2$; x is an X-axis coordinate, and y is a Y-axis coordinate; is a polynomial coefficient; and ρ_i is a normalized radial coordinate.

In this application, a surface type expression (three different surface type expressions are listed below) of the free-form lens is defined, to implement symmetry of the free-form lens in a direction of an X-axis and a direction of a Y-axis. The direction of the X-axis and the direction of the Y-axis are two directions perpendicular to each other on the imaging surface of the camera module.

In a first implementation, a surface type expression (namely, a free-form sphere curve equation) of the non-rotationally symmetric free-form lens is expressed as follows:

$$z = \frac{cr^2}{1 + \sqrt{1 - (1+k)c^2r^2}} + \sum_{i=1}^M A_i E_i$$

z is a sag of an optical surface, and z is an expression of x and y; k is a conic coefficient; c is a curvature radius; r is a radius height in the direction of the optical axis; $r^2 = x^2 + y^2$; A_i is a polynomial coefficient; and E_i is a monomial of an X-axis coordinate and a Y-axis coordinate. Directions of the x-axis and the y-axis herein are consistent with the foregoing directions of the X-axis and the Y-axis about which the free-form lens is symmetric.

$$\sum_{i=1}^N A_i E_i = A_1 x^0 y^2 + A_2 x^2 y^0 + A_3 x^2 y^2 +$$

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-continued

$$A_4x^4y^0 + A_5x^0y^4 + A_6x^6y^0 + A_7x^0y^6 + A_8x^4y^2 + A_9x^4y^2 + \dots$$

A_i is a polynomial coefficient.

Exponents of x and y in E_i of the surface type expression of the free-form lens are both even numbers, x is an X-axis coordinate, and y is a Y-axis coordinate, so that the surface type of the free-form lens has symmetry.

In a second implementation, a surface type expression of the free-form lens is:

$$z = \frac{cr^2}{1 + \sqrt{1 - (1+k)c^2r^2}} + \sum_{i=1}^M A_i E_i$$

z is a sag of an optical surface, and z is an expression of x and y ; k is a conic coefficient; c is a curvature radius; r is a radius height in the direction of the optical axis; $r^2 = x^2 + y^2$; A_i is a polynomial coefficient; and E_i is a monomial of an X-axis coordinate and a Y-axis coordinate.

$$\sum_{i=1}^N A_i E_i = A_1|x^0y^0| + A_2|x^0y^1| + A_3|x^1y^1| + A_4|x^0y^2| +$$

$$A_5|x^3y^0| + A_6|x^2y^1| + A_7|x^1y^2| + A_8|x^0y^3| + A_9|x^4y^0| + \dots$$

A_i is a polynomial coefficient, x is an X-axis coordinate, and y is a Y-axis coordinate.

In a third implementation, a surface type expression of the free-form lens is:

$$z = \frac{c_x x^2 + c_y y^2}{1 + \sqrt{1 - (1+k_x)c_x^2 x^2 - (1+k_y)c_y^2 y^2}} + \sum_{i=1}^M A_i |x^i| + \sum_{j=1}^M B_j |y^j|$$

z is a sag of an optical surface; x is an X-axis coordinate, and y is a Y-axis coordinate; k_x and k_y are conic coefficients; c_x and c_y are curvature radii; and A_i and B_j are polynomial coefficients.

In the foregoing implementations, the free-form lens may have one surface that is a free-form surface, for example, an object-side surface or an image-side surface is a free-form surface, or both the surfaces may be free-form surfaces, that is, both the object-side surface and the image-side surface are free-form surfaces.

In an implementation, FIG. 1a is a schematic diagram of a free-form lens that is symmetric in a direction of an X-axis and in a direction of a Y-axis, an imaging surface S14 is a rectangular region, and the X-axis and the Y-axis are two central axes perpendicular to each other on the imaging surface S14. An intersection of the X-axis and the Y-axis is located on an optical axis. Specifically, the direction of the X-axis is a central axis that passes through a center of the rectangular imaging surface S14 and is parallel to a long side of the rectangular imaging surface S14. The direction of the Y-axis is a central axis that passes through the center of the rectangular imaging surface S14 and is parallel to a short side of the rectangular imaging surface S14. The X-axis and the optical axis form a first plane, and the Y-axis and the optical axis form a second plane. The free-form lens L6 is a centrosymmetric structure by using the first plane as a center, and the free-form lens is also a centrosymmetric

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structure by using the second plane as a center. Keeping the free-form lens L6 centrosymmetric with respect to the first plane and centrosymmetric with respect to the second plane helps ensure imaging quality, and helps implement that imaging quality of a middle region close to the optical axis is better than imaging quality of an edge region away from the optical axis.

The following describes this application in detail by using three specific embodiments.

Embodiment 1

As shown in FIG. 2a and FIG. 2b, a straight line in the middle represents an optical axis, a left side of a camera module is an object side, and a right side of the camera module is an image side. In the camera module provided in this embodiment, a first lens L1, a stop STO, a second lens L2, a third lens L3, a fourth lens L4, a fifth lens L5, a sixth lens L6, an infrared filter element IRCF, and an electronic image sensor are sequentially arranged along the optical axis from the object side to the image side. The electronic image sensor may be placed at a position of an imaging surface S14. In this implementation, the stop STO is placed after the first lens L1, and is close to a middle position of the camera module, to help balance an aberration of the camera module.

The first lens L1 has a positive refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface S1 of the first lens L1 is convex. A region, near the optical axis, of an image-side surface S2 of the first lens L1 is convex. Both the regions are aspheric surfaces.

The second lens L2 has a positive refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface S3 of the second lens L2 is convex. A region, near the optical axis, of an image-side surface S4 of the second lens L2 is concave. Both the regions are aspheric surfaces.

The third lens L3 has a negative refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface S5 of the third lens L3 is concave. A region, near the optical axis, of an image-side surface S6 of the third lens L3 is concave. Both the regions are aspheric surfaces.

The fourth lens L4 has a positive refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface S7 of the fourth lens L4 is concave. A region, near the optical axis, of an image-side surface S8 of the fourth lens L4 is concave. Both the regions are aspheric surfaces.

The fifth lens L5 has a negative refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface S9 of the fifth lens L5 is concave. A region, near the optical axis, of an image-side surface S10 of the fifth lens L5 is concave. Both the regions are aspheric surfaces.

The sixth lens L6 has a negative refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface S11 of the sixth lens L6 is concave. A region, near the optical axis, of an image-side surface S12 of the sixth lens L6 is convex. Both the regions are free-form surfaces.

An object-side surface S13 and an image-side surface S14 of the infrared filter element IRCF are both flat surfaces.

In Embodiment 1, a distance on the optical axis between the object-side surface S1 of the first lens L1 and an imaging surface S14 of an infinitely far-away object is TTL, and an effective focal length of the camera module is EFL. The

following condition can be met: $TTL/EFL \leq 2.0$, to implement shorter TTL, thereby facilitating miniaturization design of the camera module, saving internal space of a terminal device, and facilitating thinning development of the terminal device.

In Embodiment 1, the exponents of x and y in E_i of the surface type expression of the free-form surface of the sixth lens L6 are both even numbers. This can make the surface type of the lens better symmetric, and facilitate lens processing and detection.

In an extension of Embodiment 1, optionally, a vignetting stop ST1 (not shown) may be disposed before (namely, on the object side of) the first lens L1, and a vignetting stop ST2

(not shown) may be disposed after (on the image side of) the sixth lens L6, to effectively reduce a diameter of the camera module.

In Embodiment 1, a focal length of the camera module is f , a curvature radius of the object-side surface S11 of the sixth lens L6 is R61, and a curvature radius of the image-side surface S12 of the sixth lens L6 is R62. The following condition is met: $|f/R61| + |f/R62| = 1.37$, to help correct a comprehensive aberration of a camera set, so that a lateral chromatic aberration of the camera module is less than λ μm and a distortion is less than 2%.

Table 1a is a table showing characteristics of an optical system in this embodiment. A curvature radius and a thickness are both expressed in millimeters (mm).

TABLE 1a

Surface number	Surface type	Curvature radius	Thickness	Material	Refractive index	Dispersion coefficient
S1	Aspheric surface	5.5545	0.2145	Resin	1.65	22.9
S2	Aspheric surface	43.3006	0.3991			
STO	Flat surface	Infinite	0.0856			
S3	Aspheric surface	72.4020	0.8450	Resin	1.54	56
S4	Aspheric surface	-2.2277	0.5156			
S5	Aspheric surface	-4.1395	0.2371	Resin	1.65	22.9
S6	Aspheric surface	-6.0243	0.1468			
S7	Aspheric surface	-12.8096	1.3511	Resin	1.54	56
S8	Aspheric surface	-1.5405	0.0811			
S9	Aspheric surface	-1.3990	0.2261	Resin	1.65	22.9
S10	Aspheric surface	-1.7201	0.0691			
S11	Extended aspheric surface	-3.2543	0.5182	Resin	1.65	22.9
S12	Extended aspheric surface	0.5324	1.2303			
S13	Flat surface	Infinite	0.2184	Glass	1.52	54.5
	Flat surface	Infinite	0.0520			
S14	Flat surface	Infinite	0.0000			

Table 1b gives conic coefficients k and polynomial coefficients $a1, a2, a3, a4, a5, a6, a7, a8, a9, a10, a11, a12, a13, a14$, and $a15$ that can be used for aspheric lenses surfaces S1 to S10 in Embodiment 1.

TABLE 1b

Parameter	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
K	-89.0714	130.6859	-4655.9692	-0.5553	6.5588	0.9185	45.9472	-0.7741	-0.6498	-0.4395
a1	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
a2	-0.0152	0.2774	0.1147	-0.0006	-0.1417	0.1306	-0.1796	-0.7795	-0.8846	-0.6002
a3	0.3160	0.3257	-0.0143	0.0382	-0.0718	-0.1298	-0.0613	0.2977	0.7433	0.6135
a4	0.1162	-0.2224	-0.1036	-0.7000	-0.1193	0.0432	0.1270	0.4062	0.6841	0.3106
a5	1.3304	1.3382	-0.0501	0.5297	-0.1721	0.1261	-0.3324	0.2381	0.0375	-0.3944
a6	-0.2317	3.3063	-0.0793	-0.6215	-0.1170	0.0307	-0.2253	-0.0309	-0.0484	-0.0173
a7	-1.3109	-0.8758	-0.4983	-0.1927	-0.0370	-0.0910	0.2503	-0.2238	-0.0885	0.0731
a8	0.6204	-11.8832	-0.4727	-0.3596	-0.0438	-0.1312	0.7093	-0.2444	-0.2185	0.1842
a9	-0.8465	-2.1909	0.3526	-1.7055	-0.1584	-0.1027	0.2254	-0.1653	0.0489	0.0459
a10	-0.6141	41.7733	-4.2841	-0.1863	-0.2957	-0.0327	-0.6322	-0.0264	0.1733	-0.0481
a11	3.1749	-7.4454	1.0910	-1.6257	-0.3721	0.0085	-0.1870	0.0887	0.0527	0.0010

TABLE 1b-continued

Parameter	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
a12	5.3311	-45.5026	4.8330	-4.4567	-0.2846	0.0541	-0.1634	0.1337	0.0051	-0.0070
a13	3.6623	33.6188	-0.5357	15.8836	0.0727	0.0683	0.0005	0.1275	-0.0144	-0.0087
a14	-0.9079	85.3587	15.6750	19.4730	0.7811	0.0624	0.2713	0.0584	-0.0541	-0.0043
a15	-12.2109	97.2852	-157.7664	-8.8749	1.9354	0.0098	0.4500	-0.0086	-0.0277	0.0034

Table 1c gives conic coefficients k and higher-order term coefficients X2Y0, X0Y2, X4Y0, X2Y2, X0Y4, X6Y0, X4Y2, X2Y4, X0Y6, X8Y0, X6Y2, X4Y4, X2Y6, and X0Y8 that can be used for the free-form surfaces S11 and S12 in Embodiment 1.

TABLE 1c

Parameter	S11	S12
K	-0.2054	-1.4469
X2Y0	1.7232	-1.0737
X0Y2	1.7480	-1.0137
X4Y0	-1.5261	0.0639
X2Y2	-3.0799	-0.0764
X0Y4	-1.4951	-0.0559
X6Y0	1.1911	-0.0397
X4Y2	3.4068	0.2676
X2Y4	3.0357	0.3088
X0Y6	0.8664	0.0968
X8Y0	-0.7572	0.0372
X6Y2	-2.6574	-0.1552
X4Y4	-2.2974	-0.2318
X2Y6	-1.3632	-0.2295
X0Y8	-0.1705	-0.0414

FIG. 2C shows a distortion curve of the optical system in Embodiment 1, which represents distortion values corresponding to different fields of view.

FIG. 2d shows a lateral chromatic aberration curve of the optical system in Embodiment 1, which represents lateral chromatic aberration values corresponding to five different wavelengths of light at different fields of view. Arrow indication lines are used to represent the five different wavelengths of light. The wavelengths are 510 nanometers, 470 nanometers, 610 nanometers, 550 nanometers, and 650 nanometers, respectively.

It can be learned from FIG. 2c and FIG. 2d that the optical system provided in Embodiment 1 can achieve good imaging quality.

Embodiment 2

As shown in FIG. 3a and FIG. 3b, in a camera module in this implementation, a first lens L1, a second lens L2, a stop STO, a third lens L3, a fourth lens L4, and a fifth lens L5, a sixth lens L6, an infrared filter element IRCF, and an electronic image sensor are sequentially arranged along an optical axis from an object side to an image side. The electronic image sensor can be placed on a position of an imaging surface S14 (also referred to as an image surface). In this implementation, the stop STO is placed after the second lens L2, and is close to a middle position of the camera module, to help balance an aberration of the camera module.

The first lens L1 has a positive refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface S1 of the first lens L1 is concave. A region, near the optical axis, of an image-side surface S2 of the first lens L1 is convex. Both the regions are aspheric surfaces.

The second lens L2 has a negative refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface S3 of the second lens L2 is convex. A region, near the optical axis, of an image-side surface S4 of the second lens L2 is convex. Both the regions are aspheric surfaces.

The third lens L3 has a positive refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface S5 of the third lens L3 is convex. A region, near the optical axis, of an image-side surface S6 of the third lens L3 is concave. Both the regions are aspheric surfaces.

The fourth lens L4 has a positive refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface S7 of the fourth lens L4 is concave. A region, near the optical axis, of an image-side surface S8 of the fourth lens L4 is concave. Both the regions are aspheric surfaces.

The fifth lens L5 has a negative refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface S9 of the fifth lens L5 is concave. A region, near the optical axis, of an image-side surface S10 of the fifth lens L5 is concave. Both the regions are aspheric surfaces.

The sixth lens L6 has a negative refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface S11 of the sixth lens L6 is concave. A region, near the optical axis, of an image-side surface S12 of the sixth lens L6 is convex. Both the regions are free-form surfaces.

In Embodiment 2, a distance on the optical axis between the object-side surface S1 of the first lens L1 and an imaging surface S14 of an infinitely far-away object is TTL, and an effective focal length of the camera module is EFL. The following condition can be met: $TTL/EFL \leq 2$, to implement shorter TTL, thereby facilitating miniaturization design of the camera module, saving internal space of a terminal device, and facilitating thinning development of the terminal device.

In Embodiment 2, the exponents of x and y in E_i of the surface type expression of the free-form surface of the sixth lens L6 are both even numbers. This can make the surface type of the lens better symmetric, and facilitate lens processing and detection.

In an extension of Embodiment 2, optionally, a vignetting stop ST1 (not shown) may be disposed before (namely, on the object side of) the first lens L1, and a vignetting stop ST2 (not shown) may be disposed after (on the image side of) the sixth lens L6, to effectively reduce a diameter of the camera module.

In Embodiment 2, a focal length of the camera module is f, a curvature radius of the object-side surface S11 of the sixth lens L6 is R61, and a curvature radius of the image-side surface S12 of the sixth lens L6 is R62. The following condition is met: $|f/R61| + |f/R62| = 0.71$, to help correct a comprehensive aberration of a camera set, so that a lateral chromatic aberration of the camera module is less than 3 μm and a distortion is less than 2%.

Table 2a is a table showing characteristics of an optical system in this embodiment. A curvature radius and a thickness are both expressed in millimeters (mm).

TABLE 2a

Surface number	Surface type	Curvature radius	Thickness	Material	Refractive index	Dispersion coefficient	Material
S1	Aspheric surface	-2.3847	0.3458	Resin	1.65	22.9	EP7000
S2	Aspheric surface	1.8127	0.0363				
S3	Aspheric surface	1.458	0.2855	Resin	1.67	19.243	EP9000
S4	Aspheric surface	3.0075	0.5178				
STO	Flat surface	Infinite	-0.0091				
S5	Aspheric surface	47.486	0.7245	Resin	1.54	55.99	APL5014CL
S6	Aspheric surface	-2.3808	0.6004				
S7	Aspheric surface	-7.4388	0.9962	Resin	1.54	55.99	APL5014CL
S8	Aspheric surface	-1.4872	0.096				
S9	Aspheric surface	-1.0653	0.2928	Resin	1.67	19.243	EP9000
S10	Aspheric surface	-1.6155	0.0363				
S11	Extended aspheric surface	-0.2474	0.9056	Resin	1.65	22.9	EP7000
S12	Extended aspheric surface	1.6942	1.2628				
S13	Flat surface	Infinite	0.2096	Glass	1.52	54.5	D263T
	Flat surface	Infinite	0.0499				
S14	Flat surface	Infinite	0				

Table 2b gives conic coefficients k and polynomial coefficients a1, a2, a3, a4, a5, a6, a7, a8, a9, a10, a11, a12, a13, a14, and a15 that can be used for aspheric lenses surfaces S1⁴⁰ to S10 in Embodiment 2.

TABLE 2b

Parameter	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
K	-20.2834	-2.3818	-6.3801	-57.3023	-100.0000	0.3347	23.0097	-0.5407	-0.6057	-0.7241
a1	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
a2	0.3807	-1.1425	-0.0770	0.6250	0.1903	0.1292	-0.4347	-0.8528	-0.9073	-0.5643
a3	0.0540	1.0089	0.8594	0.2524	0.0154	0.0458	0.2969	1.1047	1.4486	0.3965
a4	0.2938	-0.0461	-0.9054	-0.1618	-0.1345	-0.4954	-0.5458	-0.2717	0.7216	0.8305
a5	-0.0688	-0.0713	0.6027	-0.0884	0.0204	0.6727	-0.1883	0.1518	-0.3589	-0.5759
a6	-0.0549	-0.3815	-0.8218	1.7311	0.0268	-0.9452	0.2630	0.0884	-0.0851	-0.2502
a7	-0.0334	0.3001	-1.0009	-3.0152	-0.3675	-0.5039	0.3214	-0.1407	-0.1323	0.0253
a8	0.0225	0.3473	1.0333	-8.5801	-0.5188	1.2010	0.5801	-0.3391	-0.1651	0.2075
a9	0.0821	-0.3709	-0.5461	-0.0307	0.4198	0.1419	0.0467	-0.2976	-0.0121	0.1278
a10	0.0606	0.1619	-0.6811	47.1165	-3.5227	0.0874	-0.7888	-0.1074	-0.0245	0.0155
a11	-0.0499	-0.1433	1.7402	-1.9290	3.3752	-5.1732	-0.3589	-0.0821	-0.0951	0.0291
a12	-0.0466	-0.0746	1.4197	-54.5390	10.0663	-10.3682	-0.1919	-0.0458	-0.0297	-0.0061
a13	-0.0083	0.0858	1.4019	-16.3711	10.5950	13.9387	-0.3328	-0.1357	-0.0921	-0.0237
a14	0.0474	-0.2141	-0.1854	-50.7892	-12.5135	25.7124	0.5221	-0.0465	0.3056	-0.0251
a15	0.0322	-0.2401	-1.3685	15.2032	-156.0708	9.4206	0.7632	0.2883	0.4513	-0.0132
a16	-0.0819	0.4074	-2.0867	193.8807	170.3787	-57.1594	1.0040	0.9630	0.3100	0.0046

Table 2C gives conic coefficients k and higher-order term coefficients X2Y0, X0Y2, X4Y0, X2Y2, X0Y4, X6Y0, X4Y2, X2Y4, X0Y6, X8Y0, X6Y2, X4Y4, X2Y6, and X0Y8 that can be used for the free-form surfaces S11 and S12 in Embodiment 2.

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TABLE 2c

Parameter	S11	S12
K	-1.0178	-4.3499
X2Y0	6.5795	0.0633
X0Y2	6.5725	0.0489
X4Y0	-1.725	-0.2768
X2Y2	-3.4771	-0.5611
X0Y4	-1.7136	-0.2535
X6Y0	0.4593	0.1003
X4Y2	1.6453	0.4314
X2Y4	1.527	0.3351
X0Y6	0.4794	0.0951
X8Y0	0.1257	0.0157
X6Y2	-0.2977	-0.2672
X4Y4	0.028	-0.1712
X2Y6	-0.0265	-0.1196
X0Y8	0.0337	-0.0193

FIG. 3c shows a distortion curve of the optical system in Embodiment 2, which represents distortion values corresponding to different fields of view.

FIG. 3d shows a lateral chromatic aberration curve of the optical system in Embodiment 2, which represents lateral chromatic aberration values corresponding to five different wavelengths of light at different fields of view. Arrow indication lines are used to represent the five different wavelengths of light. The wavelengths are 510 nanometers, 470 nanometers, 610 nanometers, 550 nanometers, and 650 nanometers, respectively.

It can be learned from FIG. 3c and FIG. 3d that the optical system provided in Embodiment 2 can achieve good imaging quality.

Embodiment 3

As shown in FIG. 4a and FIG. 4b, in a camera module in this implementation, a first lens L1, a second lens L2, a stop STO, a third lens L3, a fourth lens L4, and a fifth lens L5, a sixth lens L6, an infrared filter element IRCF, and an electronic image sensor are sequentially arranged along an optical axis from an object side to an image side. The electronic image sensor can be placed on a position of an imaging surface S14.

The first lens L1 has a positive refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface S1 of the first lens L1 is concave. A region, near the optical axis, of an image-side surface S2 of the first lens L1 is convex. Both the regions are aspheric surfaces.

The second lens L2 has a negative refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface S3 of the second lens L2 is convex. A region, near the optical axis, of an image-side surface S4 of the second lens L2 is convex. Both the regions are aspheric surfaces.

The third lens L3 has a positive refractive power and is made of a resin material. A region, near the optical axis, of

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an object-side surface S5 of the third lens L3 is convex. A region, near the optical axis, of an image-side surface S6 of the third lens L3 is concave. Both the regions are aspheric surfaces.

The fourth lens L4 has a positive refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface S7 of the fourth lens L4 is concave. A region, near the optical axis, of an image-side surface S8 of the fourth lens L4 is concave. Both the regions are aspheric surfaces.

The fifth lens L5 has a negative refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface S9 of the fifth lens L5 is concave. A region, near the optical axis, of an image-side surface S10 of the fifth lens L5 is concave. Both the regions are aspheric surfaces.

The sixth lens L6 has a negative refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface S11 of the sixth lens L6 is concave. A region, near the optical axis, of an image-side surface S12 of the sixth lens L6 is convex. Both the regions are free-form surfaces.

In Embodiment 3, a distance on the optical axis between the object-side surface S1 of the first lens L1 and an imaging surface S14 of an infinitely far-away object is TTL, and an effective focal length of the camera module is EFL. The following condition can be met: $TTL/EFL \leq 2$, to implement shorter TTL, thereby facilitating miniaturization design of the camera module, saving internal space of a terminal device, and facilitating thinning development of the terminal device.

In Embodiment 3, the exponents of x and y in E_i of the surface type expression of the free-form surface of the sixth lens L6 are both even numbers, which makes the surface type of the lens better symmetric, and facilitates lens processing and detection.

In an extension of Embodiment 3, optionally, a vignetting stop ST1 (not shown) may be disposed before (namely, on the object side of) the first lens L1, and a vignetting stop ST2 (not shown) may be disposed after (on the image side of) the sixth lens L6, to effectively reduce a diameter of the camera module.

In Embodiment 3, a focal length of the camera module is f, a curvature radius of the object-side surface S11 of the sixth lens L6 is R61, and a curvature radius of the image-side surface S12 of the sixth lens L6 is R62. The following condition is met: $|f/R61| + |f/R62| = 1.92$, to help correct a comprehensive aberration of a camera set, so that a lateral chromatic aberration of the camera module is less than 3 μm and a distortion is less than 2%.

Table 3a is a table showing characteristics of an optical system in this embodiment. A curvature radius and a thickness are both expressed in millimeters (mm).

TABLE 3a

Surface number	Surface type	Curvature radius	Thickness	Material	Refractive index	Dispersion coefficient	Material
S1	Aspheric surface	-3.1589	0.2953	Resin	1.65	22.9	EP7000
S2	Aspheric surface	3.5364	0.0237				
S3	Aspheric surface	1.7330	0.2483	Resin	1.65	22.9	EP7000

TABLE 3a-continued

Surface number	Surface type	Curvature radius	Thickness	Material	Refractive index	Dispersion coefficient	Material
S4	Aspheric surface	3.3108	0.4889				
STO	Flat surface	Infinite	0.0136				
S5	Aspheric surface	19.8218	0.4821	Resin	1.54	56	APL5014CL
S6	Aspheric surface	-2.5251	0.7337				
S7	Aspheric surface	-10.0889	1.3051	Resin	1.54	56	APL5014CL
S8	Aspheric surface	-1.5229	0.0760				
S9	Aspheric surface	-1.4148	0.2782	Resin	1.65	22.9	EP7000
S10	Aspheric surface	-1.9785	0.0500				
S11	Extended aspheric surface	-2.2632	0.7622	Resin	1.65	22.9	EP7000
S12	Extended aspheric surface	0.8529	1.0613				
S13	Flat surface	Infinite	0.2100	Glass	1.52	54.5	D263T
	Flat surface	Infinite	0.0500				
S14	Flat surface	Infinite	0.0000				

Table 3b gives conic coefficients k and polynomial coefficients a1, a2, a3, a4, a5, a6, a7, a8, a9, a10, a11, a12, a13, a14, and a15 that can be used for aspheric lenses surfaces S1 to S10 in Embodiment 3.

TABLE 3b

Parameter	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
K	-19.1982	-6.7071	-10.5697	-94.8320	-92.1353	-1.8945	39.5657	-0.6213	-0.6096	-0.5025
a1	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
a2	0.0956	-0.9439	-0.1502	0.4913	0.0435	0.1196	-0.1316	-0.3738	-0.6654	-0.5632
a3	0.1720	0.6845	0.5477	0.2394	0.0128	0.0550	-0.0684	0.1254	0.8416	0.6787
a4	0.0986	0.1394	-0.5466	-0.6546	-0.0759	-0.5139	0.2599	0.3174	0.5371	-0.0965
a5	-0.1024	-0.0329	0.8873	0.4402	0.0738	0.4193	-0.3706	0.5159	-0.4902	-0.4410
a6	-0.0022	-0.5623	-0.1384	2.7731	0.0413	-0.6119	-0.1841	0.0359	-0.1847	-0.0280
a7	-0.0098	0.1792	-0.9765	0.4146	-0.4024	0.2886	0.2694	-0.3266	0.0420	0.0990
a8	-0.0333	0.3229	0.6770	-9.6616	-0.2291	0.4303	0.6533	-0.3749	-0.0774	0.2199
a9	0.0279	-0.2649	-1.2748	-1.5490	0.6105	-1.1976	0.1466	-0.2612	0.0200	0.0806
a10	0.0284	0.2324	-1.1899	32.7032	-2.3182	-0.8407	-0.6587	-0.0882	0.0473	-0.0266
a11	-0.0325	-0.1094	1.5365	-15.6447	3.9257	-3.4558	-0.2067	0.0396	-0.0928	0.0011
a12	-0.0176	-0.1524	2.5773	-61.9447	7.7852	-5.5869	-0.1438	0.0833	-0.0936	-0.0157
a13	0.0005	0.1188	1.7996	17.9166	-0.9105	15.6206	0.1141	0.0908	-0.0681	-0.0193
a14	0.0203	0.0183	-0.9110	88.2484	7.6825	24.6383	0.3079	0.0386	-0.0172	-0.0141
a15	0.0196	0.0024	-2.7400	131.1438	-159.2654	2.0639	0.4387	0.0331	0.0679	-0.0040
a16	-0.0192	-0.0171	-1.9357	-227.7872	166.8148	-66.9990	-0.4919	0.0731	0.1369	0.0094

Table 3c gives conic coefficients k and higher-order term coefficients X2Y0, X0Y2, X4Y0, X2Y2, X0Y4, X6Y0, X4Y2, X2Y4, X0Y6, X8Y0, X6Y2, X4Y4, X2Y6, and X0Y8 that can be used for the free-form surfaces S11 and S12 in Embodiment 3.

TABLE 3c

Parameter	S11	S12
K	-0.7265	-2.3120
X2Y0	1.2459	-0.4049
X0Y2	1.2611	-0.3527
X4Y0	-1.2282	-0.1014

TABLE 3c-continued

Parameter	S11	S12
X2Y2	-2.5043	-0.4262
X0Y4	-1.1911	-0.1977
X6Y0	0.9985	0.0226
X4Y2	2.8791	0.3820
X2Y4	2.6747	0.4401
X0Y6	0.7212	0.1124
X8Y0	-0.5561	0.0280
X6Y2	-1.7777	-0.1859
X4Y4	-1.6110	-0.1588
X2Y6	-0.9485	-0.1983
X0Y8	-0.0621	-0.0308

FIG. 4C shows a distortion curve of the optical system in Embodiment 3, which represents distortion values corresponding to different fields of view.

FIG. 4d shows a lateral chromatic aberration curve of the optical system in Embodiment 3, which represents lateral chromatic aberration values corresponding to five different wavelengths of light at different fields of view. Arrow indication lines are used to represent the five different wavelengths of light. The wavelengths are 510 nanometers, 470 nanometers, 610 nanometers, 550 nanometers, and 650 nanometers, respectively.

It can be learned from FIG. 4c and FIG. 4d that the optical system provided in Embodiment 3 can achieve good imaging quality.

Example embodiments of this application are described above. It should be noted that a person of ordinary skill in the art may further make several improvements and modifications without departing from the principle of this application, and these improvements and modifications are also considered to be within the protection scope of this application.

What is claimed is:

1. A camera module, comprising:

a plurality of lenses, wherein lenses of the plurality of lenses are sequentially arranged from an object side to an image side along a direction of an optical axis, wherein at least one of the plurality of lenses is a free-form lens, wherein the free-form lens is a non-rotationally symmetric lens, wherein a first lens of the plurality of lenses is a lens of the plurality of lenses nearest the object side in a direction from the object side to the image side, wherein a distance on the optical axis between an object-side surface of the first lens and an imaging surface is TTL, wherein an effective focal length of the camera module is EFL, wherein $TTL/EFL \leq 2.0$, and wherein a lateral chromatic aberration of the camera module is less than 3 μm and a distortion is less than 2%;

wherein one or more of an object-side surface or an image-side surface of the free-form lens are free-form surfaces, and wherein a surface type expression of the free-form lens is:

$$z = \frac{c_x x^2 + c_y y^2}{1 + \sqrt{1 - (1 + k_x)c_x^2 x^2 - (1 + k_y)c_y^2 y^2}} + \sum_{i=1}^M A_i |x|^i + \sum_{i=1}^M B_i |y|^i$$

wherein z is a sag of an optical surface, wherein x is an X-axis coordinate, and y is a Y-axis coordinate, wherein k_x and k_y are conic coefficients, wherein c_x and c_y are curvature radii, and wherein A_i and B_i are polynomial coefficients.

2. The camera module according to claim 1, wherein the free-form lens is symmetric with respect to a first plane, and wherein the free-form lens is also symmetric with respect to a second plane; and

wherein the first plane is a plane comprising an X-axis and the optical axis, wherein the second plane is a plane comprising a Y-axis and the optical axis, and wherein the X-axis and the Y-axis are two central axes that are perpendicular to each other on the imaging surface of the camera module.

3. The camera module according to claim 2, wherein the X-axis and the Y-axis are each central axes passing through

a center of the imaging surface and are respectively parallel to a long side and a short side of the imaging surface.

4. The camera module according to claim 3, wherein a quantity of lenses of the plurality of lenses is N, wherein $N \geq 3$, and wherein the plurality of lenses comprise the first lens to an Nth lens sequentially arranged in the direction from the object side to the image side; and

wherein surfaces of object-side surfaces and image-side surfaces of the first lens to an (N-1)th lens are all aspheric surfaces, and wherein the Nth lens is a free-form lens.

5. The camera module according to claim 2, wherein a half of a diagonal length of an effective pixel region on the imaging surface of the camera module is ImgH, and wherein $TTL/ImgH \leq 2.0$.

6. The camera module according to claim 2, wherein an entrance pupil diameter of the camera module is EPD, and wherein $EFL/EPD \leq 2.2$.

7. The camera module according to claim 2, wherein a field of view of the camera module is FOV, wherein $FOV > 100$ degrees, and wherein $EFL < 20$ mm.

8. The camera module according to claim 1, wherein a quantity of the plurality of lenses is N, wherein three lenses that are of the plurality of lenses and that are nearest the object side are arranged sequentially in the direction from the object side to the image side and are, respectively, the first lens, a second lens, and a third lens; and

wherein the camera module further comprises a vignetting stop, and wherein the vignetting stop is disposed on an object side of the second lens or on an object side of the third lens.

9. The camera module according to claim 1, wherein a lens of the plurality of lenses adjacent to the imaging surface is the free-form lens, wherein a curvature radius of the object-side surface of the free-form lens is R61, wherein a curvature radius of an image-side surface of the free-form lens is R62, and wherein $|EFL/R61| + |EFL/R62| < 2$.

10. The camera module according to claim 1, further comprising an electronic image sensor, wherein the electronic image sensor is disposed on the imaging surface, and wherein the imaging surface of the camera module is a rectangular region that matches an image sensing area of the electronic image sensor and is not less than the image sensing area of the electronic image sensor.

11. The camera module according to claim 10, wherein a diagonal length of an image sensing surface of the electronic image sensor is not less than 5.5 mm.

12. A terminal device, comprising:

a camera module, comprising a plurality of lenses having lenses sequentially arranged from an object side to an image side along a direction of an optical axis, wherein at least one lens of the plurality of lenses is a free-form lens, wherein the free-form lens is a non-rotationally symmetric lens, wherein a first lens of the plurality of lenses is a lens of the plurality of lenses nearest the object side in a direction from the object side to the image side, wherein a distance on the optical axis between an object-side surface of the first lens and an imaging surface is TTL, wherein an effective focal length of the camera module is EFL, wherein $TTL/EFL \leq 2.0$, and wherein a lateral chromatic aberration of the camera module is less than 3 μm and a distortion is less than 2%;

wherein one or more of an object-side surface or an image-side surface of the free-form lens are free-form surfaces, and wherein a surface type expression of the free-form lens is:

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$$z = \frac{c_x x^2 + c_y y^2}{1 + \sqrt{1 - (1 + k_x)c_x^2 x^2 - (1 + k_y)c_y^2 y^2}} + \sum_{i=1}^M A_i |x|^i + \sum_{i=1}^M B_i |y|^i$$

wherein z is a sag of an optical surface, wherein x is an X-axis coordinate, and y is a Y-axis coordinate, wherein k_x and k_y are conic coefficients, wherein c_x and c_y are curvature radii, and wherein A_i and B_i are polynomial coefficients.

13. The terminal device according to claim **12**, wherein the free-form lens is symmetric with respect to a first plane, and wherein the free-form lens is symmetric with respect to a second plane; and

wherein the first plane is a plane comprising an X-axis and the optical axis, wherein the second plane is a plane comprising a Y-axis and the optical axis, and wherein the X-axis and the Y-axis are each central axes that are perpendicular to each other on the imaging surface of the camera module.

14. A device, comprising:

a plurality of lenses having lenses, including a first lens and a free-form lens, sequentially arranged from an object side to an image side along a direction of an optical axis;

wherein the free-form lens is spaced apart from the first lens, wherein the first lens of the plurality of lenses is a lens of the plurality of lenses nearest the object side in a direction from the object side to the image side, wherein a distance on the optical axis between an object-side surface of the first lens and an imaging surface is TTL, wherein an effective focal length of the device is EFL, and wherein $TTL/EFL \leq 2.0$, and wherein a lateral chromatic aberration of the device is less than $3 \mu m$ and a distortion is less than 2%; and

wherein one or more of an object-side surface or an image-side surface of the free-form lens are free-form surfaces, and wherein a surface type expression of the free-form lens is:

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$$z = \frac{c_x x^2 + c_y y^2}{1 + \sqrt{1 - (1 + k_x)c_x^2 x^2 - (1 + k_y)c_y^2 y^2}} + \sum_{i=1}^M A_i |x|^i + \sum_{i=1}^M B_i |y|^i$$

wherein z is a sag of an optical surface, wherein x is an X-axis coordinate, and y is a Y-axis coordinate, wherein k_x and k_y are conic coefficients, wherein c_x and c_y are curvature radii, and wherein A_i and B_i are polynomial coefficients.

15. The device according to claim **14**, wherein the free-form lens is non-rotationally symmetric and is symmetric with respect to a first plane and symmetric with respect to a second plane different from the first plane; and

wherein the first plane is a plane comprising an X-axis and the optical axis, wherein the second plane is a plane comprising a Y-axis and the optical axis, and wherein the X-axis and the Y-axis are two central axes that are perpendicular to each other on the imaging surface.

16. The device according to claim **15**, wherein a quantity of lenses of the plurality of lenses is N, wherein $N \geq 3$, and wherein the plurality of lenses comprise the first lens to an Nth lens sequentially arranged in the direction from the object side to the image side; and

wherein surfaces of object-side surfaces and image-side surfaces of the first lens to an (N-1)th lens are all aspheric surfaces, and wherein the Nth lens is the free-form lens.

17. The device according to claim **15**, wherein the X-axis and the Y-axis are each central axes passing through a center of the imaging surface and are respectively parallel to a long side and a short side of the imaging surface.

18. The device according to claim **14**, wherein a half of a diagonal length of an effective pixel region on the imaging surface of the device is ImgH, and wherein $TTL/ImgH \leq 2.0$.

19. The device according to claim **14**, wherein a lens of the plurality of lenses adjacent to the imaging surface is the free-form lens, wherein a curvature radius of the object-side surface of the free-form lens is R61, wherein a curvature radius of an image-side surface of the free-form lens is R62, and wherein $|EFL/R61| + |EFL/R62| < 2$.

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